Pelagia Kourti

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Skills

Strong background in Theoretical Mathematics (Topology, Differential & Symplectic Geometry, TDA tools).
Quantum Computing, Quantum Information Geometry, ZX-calculus.
Optimization: Linear and dynamic programming, stochastic operational research.
Programming/Tools: Python, Qiskit, C++, FORTRAN, GNU Octave, Mathematica, LaTeX.

MSc Thesis: Geometry of Quantum Computing

Aristotle University of Thessaloniki
Analyzed the principles of quantum computing, establishing the notions of qubit, density operator and quantum logic gates. Examined the universality of the CNOT gate, the no-cloning theorem and the threshold theorem and their implications for scalable quantum simulations and noise-resilient systems.
Explored concepts in classical and quantum information theory, incl. Fisher Information, its relation to information loss, Shannon entropy, and Kullback-Leibler divergence. Analyzed the Quantum Fisher Information and its connection to the classical case, including the role of the symmetric logarithmic derivative.
Studied the Quantum Geometric Tensor, its gauge invariance, and its decomposition into the Fubini-Study metric for quantum distance measurements, and Berry curvature, with tools from symplectic geometry and Kähler manifolds. Proved the relationship of quantum geometric tensor to the Fisher metric and the Fubini-Study hermitian metric. Studied the interacting Frobenius algebras of the ZX-calculus and the resulting Hopf algebra structure on multi-qubit Hilbert spaces. Achieved circuit optimization using the ZX-rules.

Education

2023 – June 2026: MSc in Theoretical Mathematics

Aristotle University of Thessaloniki, Greece
Thesis: Geometry of Quantum Computing.
Supervisor: Panagiotis Batakidis
Grade: 8.27/10
Courses: Algebraic Topology, Differential Geometry, Homological Algebra, Category Theory, Complex Analysis, Commutative Algebra, Riemannian Geometry.

2019 – 2023: BSc in Mathematics

Aristotle University of Thessaloniki, Greece
Thesis: Introduction to the Geometry of Special Relativity.
Supervisor: Panagiotis Batakidis
Grade: 7.98/10 (Top 5%)
Courses: Stochastic Methods in Finance, Topology, Differential Geometry, Riemannian Geometry, Geometric Group Theory, Functional Analysis, Theoretical Informatics, Group Theory, Ring Theory, Complex Analysis, Real Analysis, Measure Theory, Numerical Analysis, Differential Equations, Stochastic Processes, Probability Calculus, Stochastic Operational Research, Stochastic Strategies, Statistics, Python, Linear and Dynamic Programming.

Projects/Research

2026: Optimum Chocolates' Distribution

Personal Project
Distributed chocolate bars of varying integer lengths among children with varying integer demands, using the fewest cuts possible. The problem is NP-hard; the solution is a polynomial-time heuristic that avoids exponential search.

2022 – 2023: BSc Thesis, Introduction to the Geometry of Special Relativity

Aristotle University of Thessaloniki
Studied the geometric structure of Lorentz–Minkowski space and its role in special relativity. Analyzed relativistic transformations, isometries, and causal structure using geometric methods. Modeled spacetime trajectories and distances within a rigorous mathematical framework.

2022: Proximity Game Player

Aristotle University of Thessaloniki
Developed a player for the Proximity game using Python. The player receives a number and the current state of the game board at each turn, evaluates the available empty positions, and selects the optimal position to place the given number.

Scholarships & Awards

Oct 2023 – July 2024: I & K Samara Scholarship
Full academic scholarship for MSc studies; selected as one of two students nationwide for funding.

Dec 2019 – July 2023: COSMOTE TELEKOM Scholarship
Full academic scholarship for BSc studies; renewed three times based on academic excellence.

Experience

Sept 2019 – Present: Mathematics Tutor
Freelancer
- Designed and organized educational material adapted to students of varying levels.
- Supported students with learning difficulties, fostering both their academic development and self-confidence.

July 2022 – Aug 2022: Intern through Scholarship
COSMOTE TELEKOM, Greece
- Managed documentation for the government fiber optics installation subsidy (SFBB-FTTH project).
- Handled return mail and updated related data records.

Volunteering

2021 – Present: Blood Donor

Apr 2022 – Apr 2025: Board Member
Cultural & Athletic Association INAICh, Greece
Contributed to organizational decision-making and coordination of cultural and community activities.

Feb 2022 – Jan 2024: Cellist
Symphony Orchestra, Aristotle University of Thessaloniki
Participated as cellist in collaborative musical performances, demonstrating discipline and teamwork.

Oct 2021 – Feb 2022: Volunteer Contributor (Mathematics Content Translation)
Aristotle University of Thessaloniki (WeBWorK Project)
Translated and adapted mathematical problems into English for online educational use.

Research Interests

Construction and Optimization of Quantum Algorithms, ZX-calculus, applications of mathematics to real-world problems.

Languages

English (Fluent spoken; Experienced in scientific writing), Greek (Native), German (A2)

References

References available upon request.